IIA / WEB DESIGN

INTRODUCTION TO THE WEB

How the Internet, websites and web technologies support information services

Bachelor of Library and Information Studies | Records Management Systems

By the end of Unit 1, you should be able to…

1. Distinguish the Internet from the World Wide Web.
2. Define browser, website, webpage, URL, domain, server and hosting.
3. Describe a simple browser–server request and response.
4. Identify the different functions of HTML, CSS and JavaScript.
5. Suggest safe web services for a library or records centre.
6. Create and preview a basic HTML document.

Why does web design matter in your programme?

Library & information services

A student needs to find opening hours, catalogue access, borrowing guidance and a way to contact library staff. A usable website makes these services easier to find.

Records management services

A visitor needs public request procedures, office details, access policies and clear explanations—without confidential records being exposed.

In both contexts, good web design improves access, clarity and trust.

The Internet is not the same as the Web

INTERNET

An interconnected system of computer networks. Supports many services, including email, messaging and the World Wide Web.

WORLD WIDE WEB

A system of linked resources, webpages and applications accessed through browsers over the Internet.

The Web uses the Internet; not every Internet service is a website.

Recognise the parts of a web presence

Website

A collection of related web resources and pages.

Webpage

One particular document or page view.

Homepage

A common entry point to a website.

Hyperlink

A connection to another resource or location.

Web vocabulary: people, devices and services

Client / browser

The programme asking for and displaying resources (for example Chrome).

Web server

A service that responds to web requests and provides resources.

Domain name

A human-readable Internet name, such as example.org.

Hosting

The infrastructure/service that makes a website available.

How a browser displays a webpage



Browser requests a resource → Server responds → Browser renders HTML/CSS and may run JavaScript.

Client and server: who does what?

Client: the visitor’s browser

Accepts an address, makes a request, interprets returned files, and shows the webpage. It can display files saved directly on a computer.

Server: the website provider

Receives a request and returns content or a response. Some services also process accounts, forms and database queries.

A simplified model: real web requests may involve networks, DNS, caches and multiple servers.

Read a web address (URL)

```
https://library.example.org/catalogue/search?keyword=history
```

Protocol	Host	Path	Query
https	library.example.org	/catalogue/search	keyword=history

example.org is a reserved example domain, not the address of a real library.

Does HTTPS mean a website is safe?

HTTPS helps protect data while it travels between browser and server.

What it does

Helps encrypt communication in transit.

What it does not do

Guarantee that an organisation or webpage is honest, accurate or secure in every way.

Three web technologies, three main jobs

HTML → Structure

Headings, paragraphs, images, lists, tables, forms and links. Defines what the content is.

CSS → Presentation

Colours, type, spacing, borders, alignment and layout. Controls how content looks.

JavaScript → Behaviour

Menus, feedback, changing content and other interaction. Tells a page how to respond.

This course emphasises HTML and CSS; JavaScript is introduced through simple examples.

A small HTML document

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Library Service</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>Find library services here.</p>
</body>
</html>
```

<head>

Metadata and browser-tab title.

<body>

Visible webpage content.

<h1> / <p>

A main heading and a paragraph, with opening and closing tags.

Small HTML changes create a meaningful page

```
<h1>Library Services</h1>
<p>How can we help?</p>
<ul>
  <li>Find books</li>
  <li>Request assistance</li>
</ul>
```

Library Services

How can we help?

- Find books
- Request assistance

Edit the HTML in the unit's built-in editor or download the starter file to practise offline.

Static websites and dynamic applications

Static page / site

Typically served as files such as HTML and CSS. Appropriate for public opening hours, services, learning notes and simple project websites.

Dynamic application

May combine browser code, server-side logic, an API, authentication and databases. Appropriate for changing user data, borrowing accounts or restricted records.

A static site can still have interactive JavaScript; the difference is not simply whether buttons exist.

Where can websites be hosted?

Static hosting

Suitable for student portfolios and documentation. Example: GitHub Pages.

Shared hosting

Common for smaller websites; features and service limits vary.

Managed CMS

Suitable for staff-updated content; consider permissions and maintenance.

Cloud / virtual server

Flexible, but more responsibility for configuration, security and cost.

What makes a website good for its users?

Clear purpose

Identify user needs and important tasks.

Readable content

Use informative headings and plain language.

Good navigation

Make key information easy to locate.

Accessibility

Consider diverse devices and user abilities.

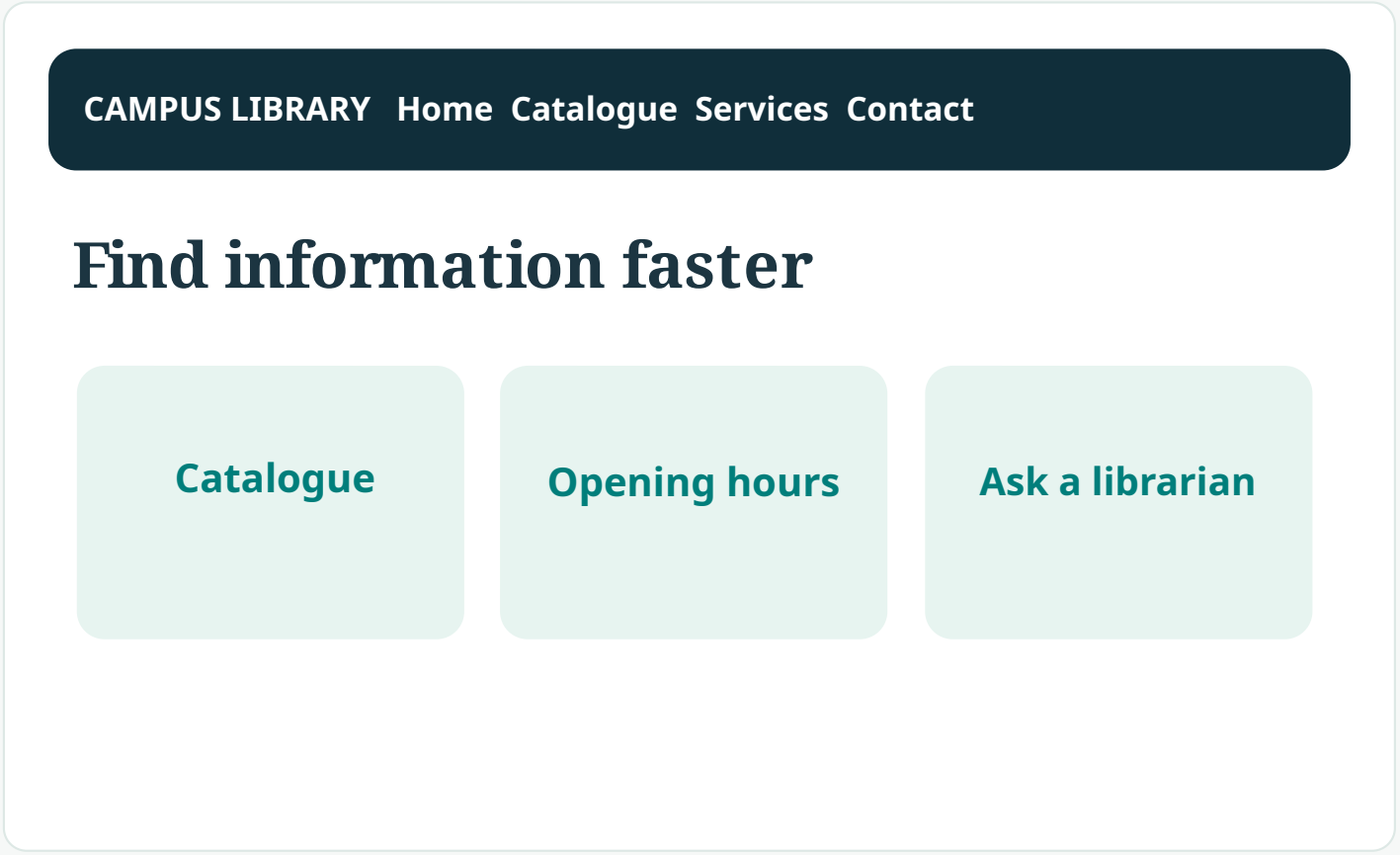
Trust and accuracy

Maintain reliable and current information.

Privacy

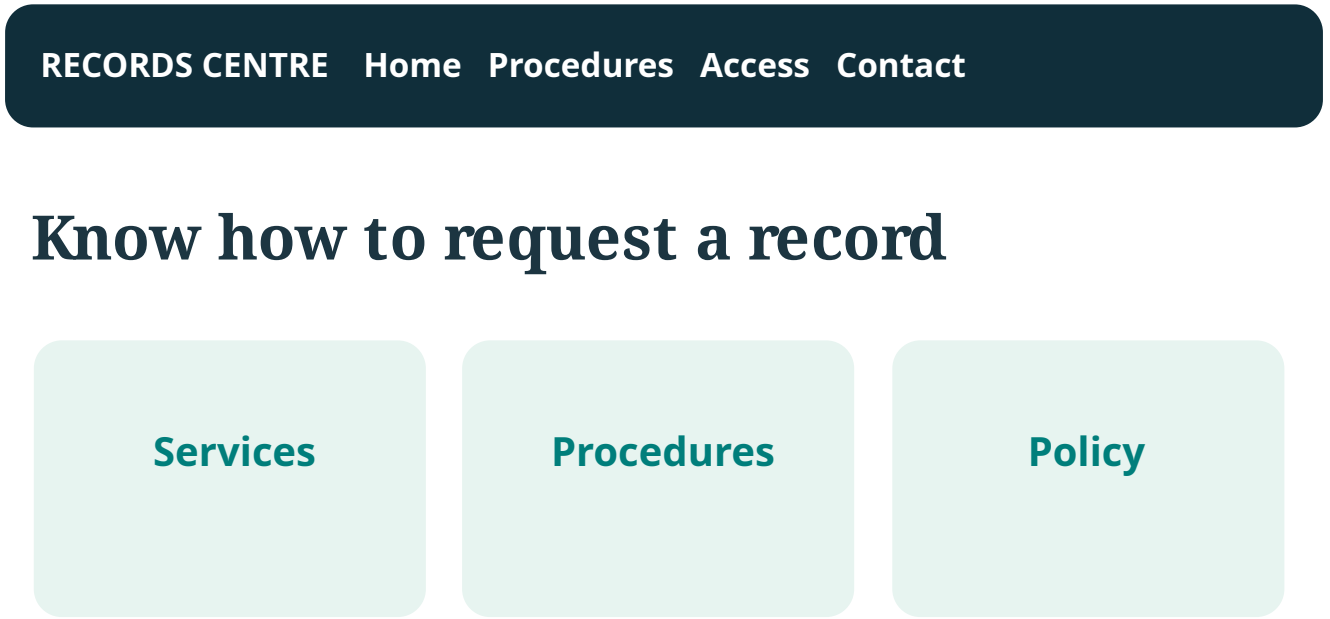
Protect confidential records and personal data.

Example: a library information website



A well-planned homepage prioritises frequent reader tasks. Useful content: catalogue access, opening hours, borrowing guidance and enquiries.

Example: a records-centre website



A public information page may explain access procedures. It must not expose protected files, identifiable records or confidential credentials.

Unit 1 guided practical

01

Inspect a public website

Identify its URL, domain, audience, navigation and one improvement.

02

Analyse a URL

Break a sample address into protocol, host, path and query.

03

Create your first HTML file

Open the starter file, change a title, heading and services list, and preview in a browser.

Assignment 1: two complementary parts

Part A: website investigation

Explain an appropriate website, intended visitors, its navigation and technical roles. Describe strengths, realistic improvements and one accessibility or privacy issue.

Part B: your first HTML page

Build a fictional library or records-centre page with title, heading, introduction, list of at least four services and one extra useful section.

Official marks, due date and Google Form will be added once approved; keep your HTML file and screenshot.

Quick check before you finish

1. Can you explain why the Web and Internet are different?
2. Can you identify a browser, a webpage and a server?
3. Can you explain HTML, CSS and JavaScript in one sentence each?
4. Can you identify the parts of a URL?
5. Can you suggest safe public website features for your profession?

Six takeaways from Unit 1

Internet ≠ Web

Webpages make up websites

A browser requests; a server responds

HTML = content structure

CSS = presentation; JavaScript = behaviour

Useful websites prioritise people and information safety

Next: Unit 2 — Principles of Good Web Design.

Further reading and independent practice

MDN Web Docs: How the Web Works

developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Web_standards/How_the_web_works

W3Schools HTML Tutorial

w3schools.com/html/

The course website

Units → Unit 1 → Practicals / Self-check / Downloads